The theoretical limit is reached when equality holds.

V.

Reference solution:

Newton's equations for the two-dimensional electrons along and are

In steady state is time-independent, so . Thus gives

Substitution into gives

By definition,

Thus a current also flows perpendicular to the electric field.

VI.

Reference solution:

The electric field between the plates is

The plates are ideal conductors, so at and , . Hence

Using the relation below, the wave in can be

expressed as the superposition of two plane electromagnetic waves: